

Geospatial Data Science & ML — Foundations

The domain, not the project: how spatial data works and how ML models it — plus things to read and listen to casually.

General-fluency primer · complements (does not repeat) the project + role reading lists · Aug 2026 · every link verified against primary sources

What this is. The other two lists were built for *this project* and the *Esri role*. This one is for the domain itself — the conceptual spine of spatial data science and how machine learning actually behaves on geographic data. Nothing here duplicates the earlier lists. §1–3 are the ideas worth genuinely internalizing; §4–6 are the modern ML/EO landscape; §7–8 are for relaxed, ongoing absorption.

If you only do three things: (1) understand **why spatial data breaks normal ML** — §1 + the spatial-cross-validation papers in §3; (2) skim the **Geographic Data Science with Python** book (§7, free); (3) follow one or two idle sources in §8 so the field stays in your peripheral vision.

1 — Why Spatial Is Different (the ideas that actually matter)

A Computer Movie Simulating Urban Growth in the Detroit Region — Waldo Tobler (1970), *Economic Geography* 46(2) [JSTOR / paywalled](#)

The origin of Tobler's First Law ("near things are more related than distant things") — the founding idea behind spatial dependence.

doi.org/10.2307/143141

What is Tobler's First Law of Geography? — GIS Geography

Plain-language explainer linking the First Law to spatial autocorrelation and Moran's I — read this if the Tobler paper is behind a wall.

gisgeography.com/tobler-first-law-of-geography/

The Modifiable Areal Unit Problem and GIS — Mark Altaweel, *Geography Realm*

Why the boundaries and scale you aggregate to can change — even reverse — your results. A trap every spatial analyst must know.

geographyrealm.com/modifiable-areal-unit-problem-gis/

Pitfalls to Avoid (Ch. 6) — Manuel Gimond, *Intro to GIS and Spatial Analysis* (Colby)

MAUP and the ecological fallacy side by side with worked aggregation examples. Free, concise.

mgimond.github.io/Spatial/06_pitfalls_to_avoid.html

Lagrange Multiplier Test Diagnostics for Spatial Dependence and Spatial Heterogeneity —

Luc Anselin (1988), *Geographical Analysis* 20(1) [Wiley / paywalled](#)

The formal distinction between spatial dependence and spatial heterogeneity — the two ways spatial data violates classical statistics.

doi.org/10.1111/j.1538-4632.1988.tb00159.x

2 — Spatial Statistics & Analysis (the classical modeling toolkit)

PySAL — Python Spatial Analysis Library — pysal developers

The umbrella for the standard open-source spatial-statistics ecosystem in Python — your entry point to all of the below.

pysal.org/pysal/

esda — Exploratory Spatial Data Analysis — pysal developers

Implements global/local Moran's I, Geary's C, Getis-Ord G, and LISA — how you detect and map spatial clusters.

pysal.org/esda/stable/

Local Spatial Autocorrelation: LISA and Local Moran — Luc Anselin, GeoDa workbook
Tutorial from the field's originator on reading local Moran's I cluster maps. The intuition, not just the code.
geodacenter.github.io/workbook/6a_local_auto/lab6a.html

Spatial Autocorrelation (Ch. 13) — Manuel Gimond, Intro to GIS and Spatial Analysis
Worked math-plus-code on global/local Moran's I, grounded in Tobler's Law.
mgimond.github.io/Spatial/13_autocorrelation.html

spreg — Spatial Regression in Python — pysal developers
Spatial econometrics: lag/error models and diagnostics — regression that respects spatial dependence.
pysal.org/spreg/

mgwr — (Multiscale) Geographically Weighted Regression — Oshan, Li, Kang, Wolf & Fotheringham
Lets a relationship vary across space (and scale) instead of assuming one global coefficient — a core spatial idea.
mgwr.readthedocs.io/en/latest/

PyKriging — Kriging Toolkit for Python — GeoStat-Framework
Standard Python geostatistics/kriging (variograms, ordinary/universal kriging) — how you interpolate spatial fields.
geostat-framework.readthedocs.io/projects/pykriging/en/stable/

3 — ML on Spatial Data — Pitfalls & Methods (read before you trust any accuracy number)

Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure — Roberts, Bahn, Ciuti et al. (2017), *Ecography* 40(8)
The foundational argument that random CV overestimates accuracy under spatial autocorrelation — and how block CV fixes it. Essential.
doi.org/10.1111/ecog.02881

Spatially autocorrelated training and validation samples inflate performance assessment of CNNs — Kattenborn et al. (2022), *ISPRS Open Journal of Photogrammetry & Remote Sensing* 5
Concrete proof in remote sensing: random holdout inflated CNN accuracy by up to 28% vs. spatial block CV. The number that makes it click.
doi.org/10.1016/j.ophoto.2022.100018

GroupKFold — scikit-learn docs
The general-purpose sklearn building block for grouped/blocked splits that prevent spatial leakage.
scikit-learn.org/stable/modules/generated/sklearn.model_selection.GroupKFold.html

Spatial cross-validation using scikit-learn — Chiara Ledesma, *Towards Data Science* (2021)
Hands-on walkthrough applying group-based splitting to geographic regions — the practical how-to.
towardsdatascience.com/spatial-cross-validation-using-scikit-learn-74cb8ffe0ab9/

mlr3spatiotempcv — Schratz, Becker, Lang & Brenning, *J. Statistical Software* 111(7) (2024)
The most complete production-grade toolkit of spatial/spatiotemporal resampling schemes (block CV, buffered LOO). The reference implementation.
mlr3spatiotempcv.mlr-org.com/

spacv — Sam Comber (GitHub)
sklearn-style Python package for grid-based and spatial leave-one-out CV — the closest Python analogue to mlr3spatiotempcv.
github.com/SamComber/spacv

Verde — Fatiando a Terra project
Gridding/interpolation with built-in spatial cross-validation and spatial feature engineering — pitfall-aware tooling in one place.
fatiando.org/verde/

4 — Deep Learning for Earth Observation (the broad field, beyond buildings)

Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources — Zhu,

Tuia, Mou, Xia, Zhang, Xu & Fraundorfer (2017), IEEE GRSM 5(4)

The canonical survey that framed the whole DL-for-remote-sensing field. Start here for the landscape.

arxiv.org/abs/1710.03959

Foundation Models for Remote Sensing and Earth Observation: A Survey — Xiao, Xuan, Wang et al. (2024)

The modern complement to Zhu et al. — vision / vision-language / LLM foundation models for EO, as of 2024.

arxiv.org/abs/2410.16602

Advances and Challenges in DL-Based Change Detection for Remote Sensing — Remote Sensing 16(5) (2024)

Recent focused survey on change detection — a core EO task with no equivalent in ordinary CV.

doi.org/10.3390/rs16050804

BigEarthNet — Sumbul, Charfuelan, Demir & Markl (2019), IGARSS

The standard large-scale multi-label Sentinel-1/2 land-cover benchmark — how EO models are compared.

arxiv.org/abs/1902.06148 · bigearth.net

EuroSAT — Helber, Bischke, Dengel & Borth (2019), IEEE JSTARS

A lightweight Sentinel-2 land-use/land-cover benchmark — the friendliest first hands-on EO dataset to actually train on.

arxiv.org/abs/1709.00029 · github.com/phe1ber/EuroSAT

LEVIR-CD — Chen & Shi (2020), Remote Sensing 12(10)

The most widely used high-res bitemporal building change-detection benchmark — the go-to for change-detection practice.

justchenhao.github.io/LEVIR/

5 — Geospatial / EO Foundation Models (the 2023–2025 frontier)

Prithvi-EO-2.0 — IBM Research + NASA IMPACT (2023–24)

The flagship open-weights EO foundation model (ViT + masked autoencoder on Harmonized Landsat-Sentinel). The field's reference point.

huggingface.co/ibm-nasa-geospatial/Prithvi-EO-2.0-300M

Clay Foundation Model — Clay Foundation (2023–24)

Fully open-source (Apache-2.0), multi-sensor embedding model (Sentinel-1/2 + DEM). The community-driven counterpart to Prithvi.

github.com/Clay-foundation/model · madewithclay.org

SatMAE — Cong, Khanna, Meng et al., Stanford (NeurIPS 2022)

Established masked-autoencoder pretraining with temporal/multi-spectral encodings as the standard recipe for satellite imagery.

arxiv.org/abs/2207.08051

Presto — Tseng et al., NASA Harvest (2023)

A deliberately lightweight pixel-timeseries transformer showing small efficient models can rival much larger EO models — the pragmatic counterpoint.

arxiv.org/abs/2304.14065

DOFA (Dynamic One-For-All) — Xiong, Wang, Zhang et al. (2024)

A single backbone that adapts to any number of input channels/sensors via a wavelength-conditioned hypernetwork — where multi-sensor EO is heading.

arxiv.org/abs/2403.15356

segment-geospatial (samgeo) — Qiusheng Wu & Lucas Prado Osco (2023, JOSS)

Applies Meta's Segment Anything Model to geospatial rasters with minimal code — the "try a foundation model in 10 lines" entry point.

github.com/opengeos/segment-geospatial

AlphaEarth Foundations — Brown et al., Google DeepMind (Jul 2025)

The newest leap: a global embedding-field model (optical+SAR+LiDAR+climate) shipped as an Earth Engine dataset — foundation models as a queryable data layer, not just a checkpoint.

arxiv.org/abs/2507.22291

6 — Geo-ML Tooling & Frameworks (the ecosystem beyond Esri)

TorchGeo — originally Microsoft, now an OSGeo project

A torchvision-style PyTorch library purpose-built for geospatial datasets, samplers, and pretrained models. The main geo deep-learning library.

github.com/torchgeo/torchgeo · docs.torchgeo.org

Raster Vision — Azavea / Element84

End-to-end, config-driven framework for training classification/detection/segmentation on large satellite/aerial imagery — how teams productionize EO ML.

github.com/azavea/raster-vision · docs.rastervision.io

Leafmap — Qiusheng Wu

Interactive Jupyter mapping (wraps ipyleaflet/folium/kepler/pydeck) with minimal code — fastest way to see your spatial data.

github.com/opengeos/leafmap · leafmap.org

eo-learn — Sinergise / Sentinel Hub

EOPatch/EOTask pipeline framework chaining preprocessing, masking, and ML over spatio-temporal satellite sequences — a look at real EO pipeline design.

github.com/sentinel-hub/eo-learn

Google Earth Engine — Google

The dominant petabyte-scale planetary geospatial compute/data platform. Essential fluency alongside — and contrast to — the Esri stack.

developers.google.com/earth-engine

geemap — Qiusheng Wu / GEE community

The standard Python bridge for driving Earth Engine interactively from Jupyter with built-in visualization.

github.com/gee-community/geemap · geemap.org

7 — Books & Longer Reads (mostly free / open)

Geographic Data Science with Python — Sergio Rey, Dani Arribas-Bel, Levi John Wolf (2020, ongoing)

The canonical free open-notebook text bridging GIS and Python data science (PySAL/GeoPandas throughout). Your single best broad read.

geographicdata.science/book/

Geocomputation with Python — Dorman, Graser, Nowosad & Lovelace (2025)

Open-source, Python-native, aimed squarely at a geospatial data scientist's daily toolstack.

py.geocompx.org/

Geocomputation with R — Lovelace, Nowosad & Muenchow (2nd ed., 2024)

Free reference for spatial data structures and vector/raster workflows whose concepts transfer directly to Python/Esri thinking.

r.geocompx.org/

Spatial Statistics for Data Science: Theory and Practice with R — Paula Moraga (2023)

Free CC-licensed text on point patterns, geostatistics, and areal models — builds the statistical intuition behind spatial ML.

paulamoraga.com/book-spatial/

Cloud-Based Remote Sensing with Google Earth Engine — Cardille, Crowley, Saah & Clinton, eds. (2024)

55-chapter open-access book from Earth Engine basics to applied remote-sensing / GeoAI use cases.
eefabook.org/

8 — Idle Learning (follow casually — keep the field in your peripheral vision)

MapScaping Podcast — MapScaping

Long-running, still-active interview podcast on the geospatial industry — good commute/walk listening.
mapscaping.com/podcast/

Modern GIS with Matt Forrest — Matt Forrest (newsletter + YouTube)

"Spatial on Saturdays" newsletter plus a big YouTube channel bridging traditional GIS and the modern/cloud data stack. Approachable and current.
mattforrest.substack.com · youtube.com/@MattForrest

Spatial Thoughts — Ujaval Gandhi

Free tutorials on QGIS, Python, and Earth Engine from a former Google Geo staffer — great for hands-on grounding.
spatialthoughts.com/

Development Seed Blog — Development Seed

Practitioner posts on open-source geospatial tooling, STAC, cloud-native EO, and applied GeoAI. How the frontier is actually built.
developmentseed.org/blog

Radiant Earth / Source Cooperative — Radiant Earth Foundation

Nonprofit behind Source Cooperative, a vendor-neutral open geospatial data-hosting platform — good pulse on open EO data.
radiant.earth · source.coop

Microsoft Planetary Computer Blog — Microsoft

Official updates on Planetary Computer datasets/APIs and applied EO/GeoAI work.
techcommunity.microsoft.com/category/azure/blog/microsoft-planetary-computer-blog

Free and Open Source GIS Ramblings — Anita Graser

Long-running blog on open-source GIS and movement/trajectory data science (MovingPandas). A different, mobility-flavored corner of the field.
anitagraser.com/

Every link verified to resolve against a primary source (Aug 2026). A few items (Tobler 1970, Anselin 1988, and some publisher/doc domains) sit behind paywalls or block automated fetchers but were confirmed real via DOI resolution and cross-checking — flagged inline where relevant. Companion documents: *Urban Rooftop Solar Potential — Project Spec* and *Geospatial ML for Solar — Reading List*.